Attendance Monitoring System

Introduction To Computing Project
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1.1 Introduction

Attendance monitoring is an important process in schools because it helps keep track of students' presence and absence. However, traditional manual attendance recording can take time and may lead to difficulties in organizing and managing attendance records. Rivera (2021), in the study Enhanced Attendance Monitoring System Using Biometric Fingerprint Recognition, presented a computerized attendance system that aimed to improve the traditional manual attendance process. Similarly, Acharya (2024), in Attendance Management System, developed a stand alone attendance software that computerizes attendance recording and automatically generates reports. These studies provide support for using computerized attendance systems to make attendance recording more organized and efficient. The project assumes a school setting where teachers record student attendance. Manual attendance recording can take time, especially when there are many students. It may also result in incorrect or difficult-to-check records. Another concern is keeping attendance organized by section and date, particularly when records become numerous. The proponents developed the Attendance Monitoring System using a Python console-based application. The main user is the teacher, while students and school administrators are intended users for viewing attendance information. The system allows users to add, view, and search students, record attendance by section and date, set Present/Absent/Late status, update records, delete student records, prevent duplicate attendance for the same student and date, and calculate attendance totals.

1.2 Statement of the Problem

General Problem

Manual attendance recording can make it difficult for teachers to record and manage student attendance efficiently. The process can take time and may lead to mistakes when attendance records are written, checked, or organized. The proposed system addresses this difficulty through a simple Python console-based attendance monitoring program.

Specific Problems

1. Recording attendance manually can take time, especially when there are many students.

2. Attendance records can contain mistakes or duplicate entries when they are recorded or checked manually.
3. Organizing and checking attendance by section and date can be difficult when records are numerous.

1.3 Objectives of the Study

General Objective

To develop an Attendance Monitoring System, a Python console-based application that enables teachers to record, update, organize, and check student attendance by section and date.

Specific Objectives

1. To provide functions for adding, viewing, and searching student information, including Student ID, name, age, and section.
2. To record and update attendance using Present, Absent, and Late statuses while preventing duplicate attendance for the same student and date.
3. To organize attendance records by section and date and calculate each student's total Present, Absent, and Late records.

1.4 Scope

The Attendance Monitoring System is a Python-based console/terminal program designed for a school setting. The teacher is the main user for recording and managing attendance, while students and school administrators are intended to view attendance information. Student and attendance information are stored in Python lists while the program is running.

- Add Student – The user can enter and save the student's Student ID, name, age, and section.
- View and Search Students – The user can view the registered student list and search for a student using the Student ID.
- Record Attendance – The user can select a section and date, then record Present, Absent, or Late status for students in that section.

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- .Manage Records – The user can view attendance by section and date, update existing attendance records, and delete student records when needed.
- Attendance Summary – The system calculates each student's total Present, Absent, and Late records. The system also checks for duplicate attendance for the same student and date.

1.5 Limitations

The system is limited to a console/terminal environment and does not include a graphical user interface. It does not provide online, web-based, or cloud features. Student and attendance information is stored in Python lists while the program is running and is not connected to a database or cloud storage. The system does not include grading, enrollment, payment, notifications, or other school-management features. The project focuses only on student attendance monitoring. The current program is intended for local use and does not support simultaneous multi-user access.

16. System Requirements

Hardware Requirements

- Laptop or desktop computer with a working processor and at least 4 GB RAM.
- At least 1 GB of available storage for the Python project and related files.
- Monitor or screen for viewing the console output.
- Keyboard and mouse for entering information and selecting program options.

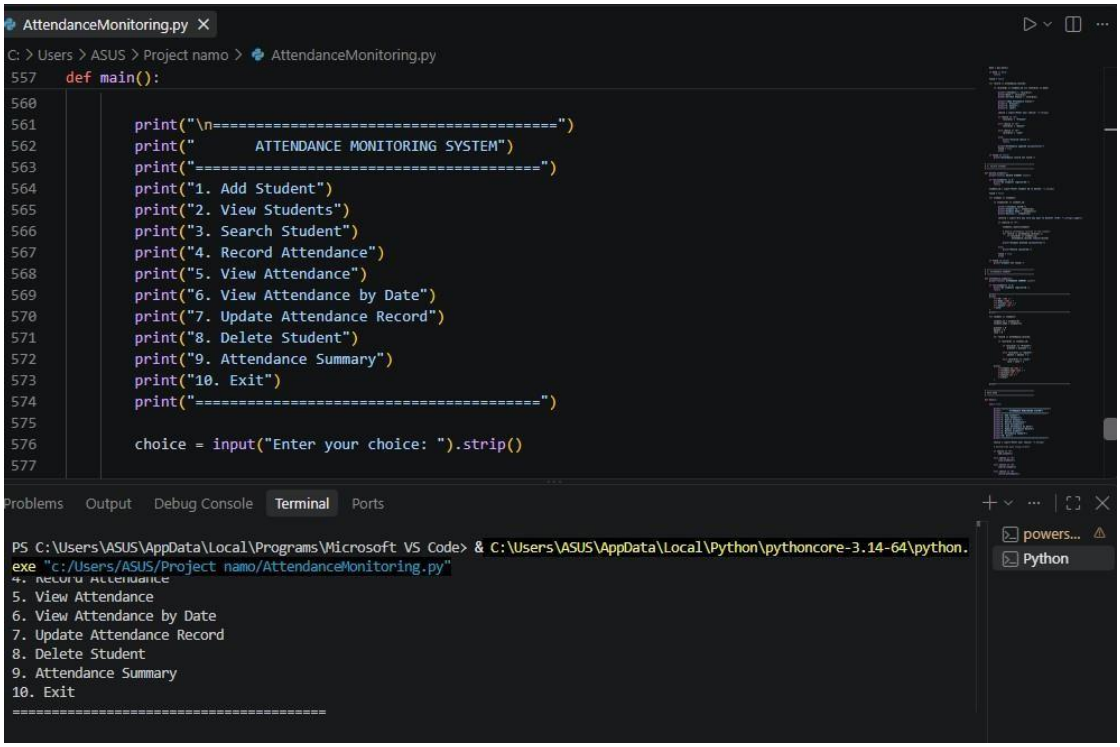
Software Requirement

- Windows 10/11 operating system.
- Python 3.13.x for developing and running the program.
- Visual Studio Code for writing and editing the Python source code.
- Command Prompt/Terminal or Python IDLE for running and testing the console program.

Functional Requirements

- The teacher can add a student by entering the Student ID, name, age, and section, and the system stores the information.
- The teacher can record attendance by section and date using the Present, Absent, or Late status.
- The system validates Student IDs and prevents duplicate student IDs and duplicate attendance for the same student and date.
- The user can search students, view attendance by section or date, update attendance, delete student records, and view attendance totals.

1.7 Code Snippet of Scheduling System for Claiming School Uniform



Snippet 1 Add Student and Duplicate Student ID Validation

```
      exists = False


for student in students:
    if student[0] == student_id:
        exists = True
        break


if exists:
    print("This student is already registered!")
    print("Please enter a different Student ID.")
else:
    students.append([student_id, student_name, age, section])
    print("Student added successfully!")
          rip()
```

This code checks for duplicate Student IDs and adds the student's information if the ID is not already registered.

Snippet 2 Record Attendance and Prevent Duplicate Attendance

```
duplicate = False

for record in attendance_records:
    if record[0] == student_id and record[2] == date:
        duplicate = True
        break

if duplicate:
    print("Attendance already recorded.")
    continue

print("1. Present")
print("2. Absent")
print("3. Late")

choice = input("Enter attendance choice: ").strip()

if choice == "1":
    status = "Present"
elif choice == "2":
    status = "Absent"
elif choice == "3":
    status = "Late"
else:
    print("Invalid choice. Attendance not recorded.")
    continue

attendance_records.append([student_id, student_name, date, status])
print("Attendance recorded successfully!")
```

This code prevents duplicate attendance for the same student and date and records the selected status: Present, Absent, or Late.

Snippet 3 Attendance Summary and Attendance Totals

```
present = 0
absent = 0
late = 0

for record in attendance_records:
    if record[0] == student_id:
        if record[3] == "Present":
            present += 1
        elif record[3] == "Absent":
            absent += 1
        elif record[3] == "Late":
            late += 1

print("Present:", present)
print("Absent:", absent)
print("Late:", late)
```

This code counts each student's total Present, Absent, and Late attendance records.

References

Acharya,K.(2024).*Attendancemanagement system*.Authorea.https://doi.org/10.22541/au.171098658.87381421/v1

Rivera, R. B. (2021). Enhanced attendance monitoring system using biometric fingerprint recognition. *International Journal of Recent Technology and Engineering, 9*(5), 1–4. https://www.ijrte.org/portfolioitem/E5070019521/